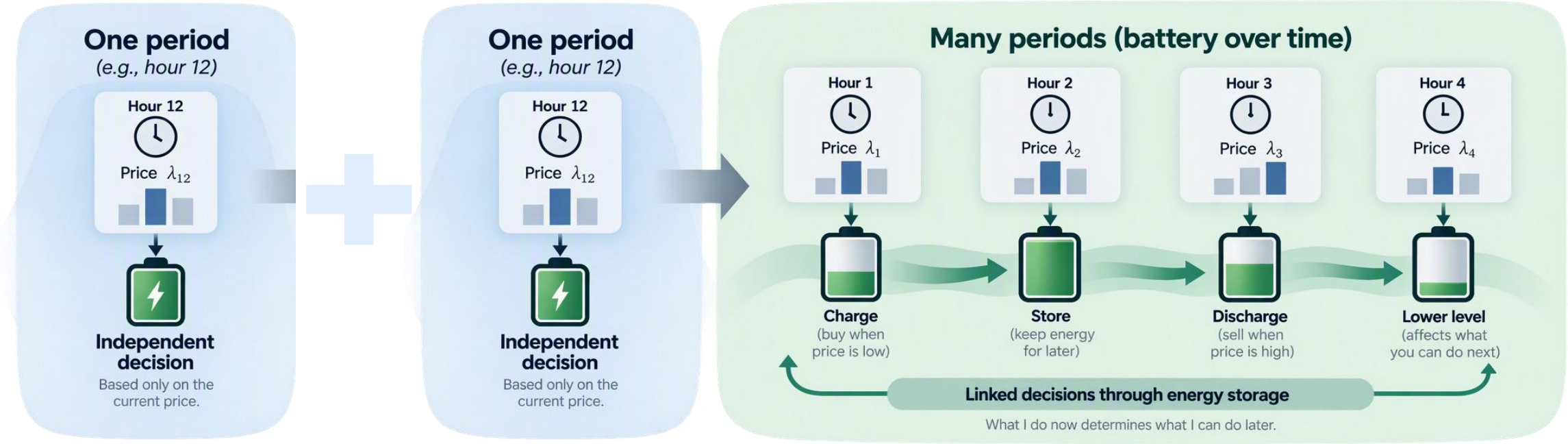
46750 – Optimization in Modern Power Systems

Multi-period optimization: The value of stored energy

Where we are

From one time period to multiple: the same duality, a new coupling



In week 4:

- Best response of self-scheduling generator, given hourly price λ
- **Opportunity cost** arises from scarce capacity

In week 5:

- Best response of self-scheduling battery, given price profile λ_t
- **Opportunity cost** arises from coupling between hours

The battery — Key characteristics



Energy capacity
$\bar{E}$ = 15 kWh

Power limit
$\bar{P}$ = 10 kW
(charge and discharge)

No losses
(for now)

Starts empty

Your 15 kWh battery is full. The price is 100 DKK/kWh, above the daily average. Do you discharge now?

- ▲ Yes - sell while the price is high
- ◆ No - wait for better prices
- ● Not enough information to answer this question

Three questions for today's lecture



Three questions for today's lecture

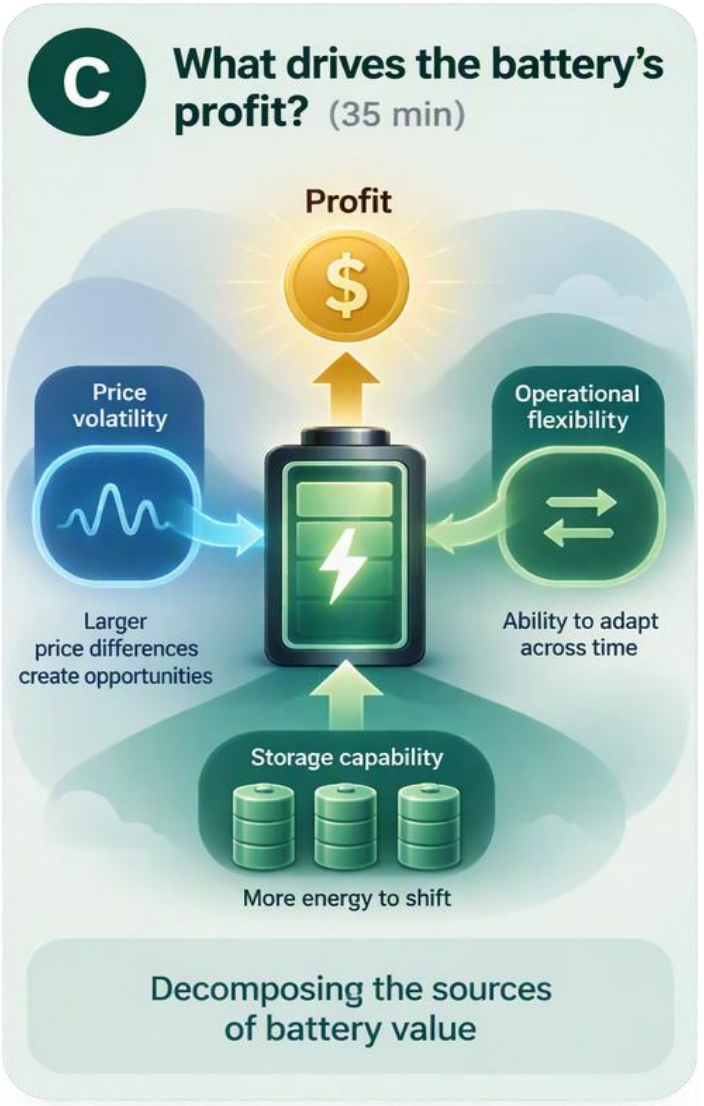




Three questions for today's lecture



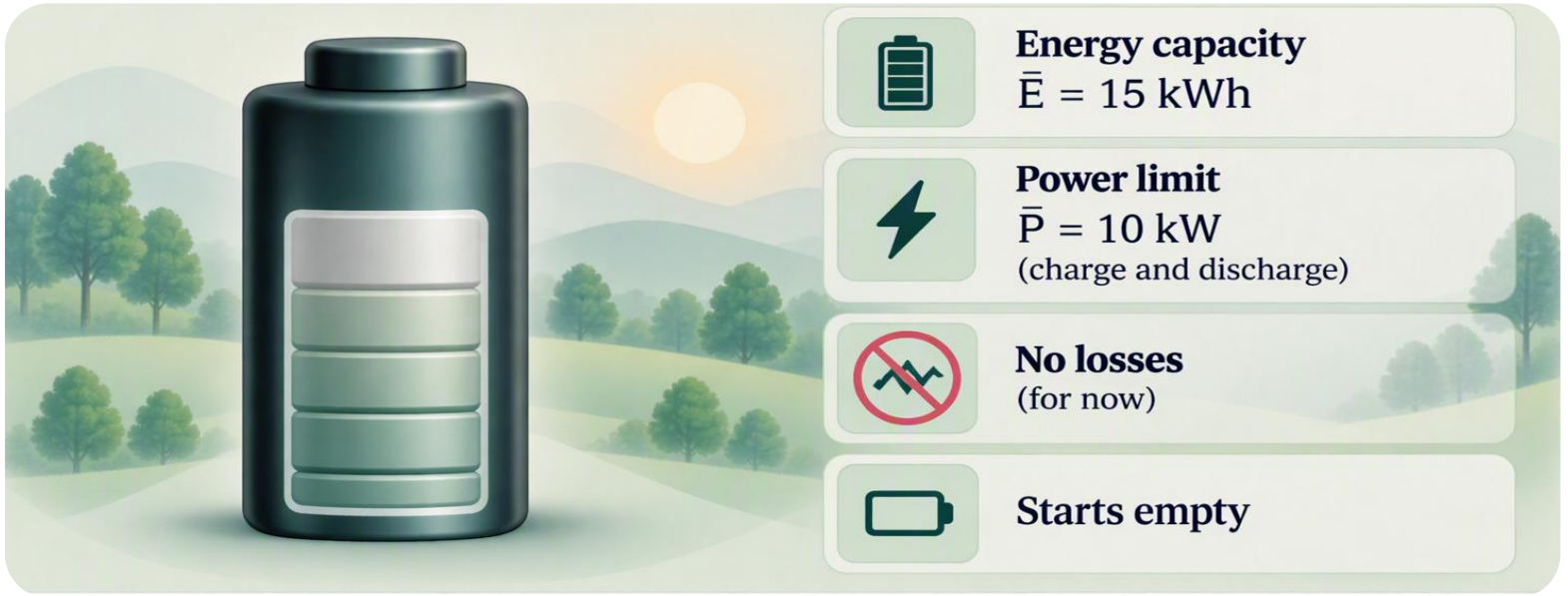


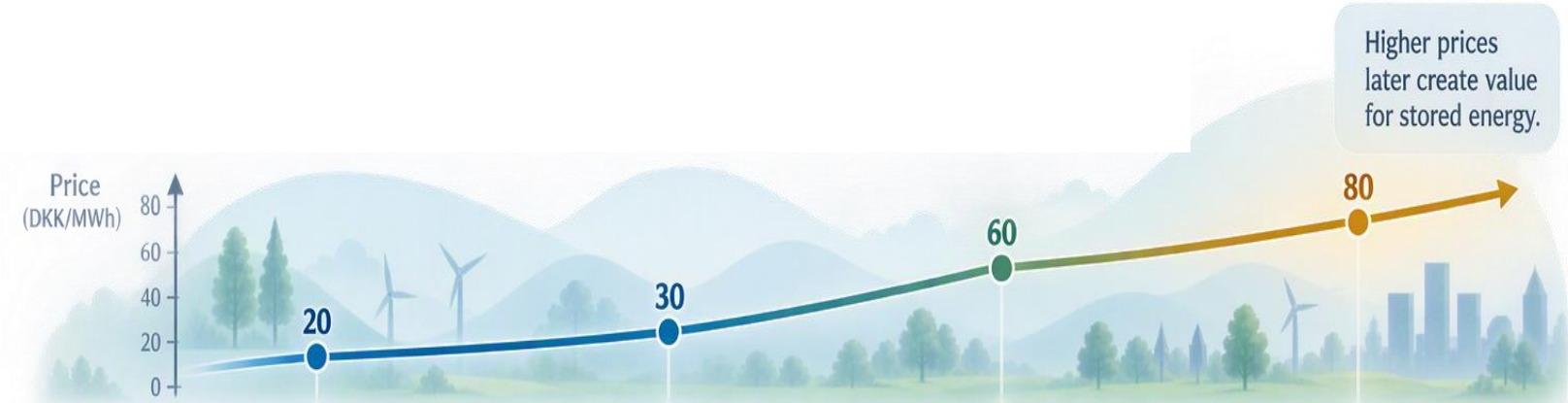


BLOCK A

When should a battery charge or discharge?

Four-hour toy example





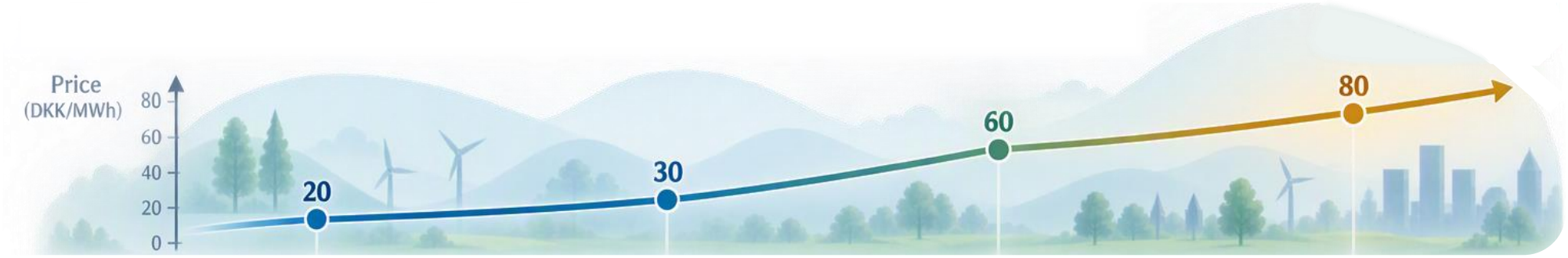


Which one of these charging/discharging profiles is optimal for the battery?

- **Profile A)** h1: + 10 kW; h2: 0 kW; h3: 0 kW; h4: + 10 kW
- **Profile B)** h1: + 10 kW; h2: - 10 kW; h3: + 10 kW; h4: - 10 kW
- **Profile C)** h1: + 10kW; h2: + 5 kW; h3: - 10 kW; h4: - 5 kW
- **Profile D)** h1: + 10 kW; h2: + 5 kW; h3: - 5 kW; h4: - 10 kW

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



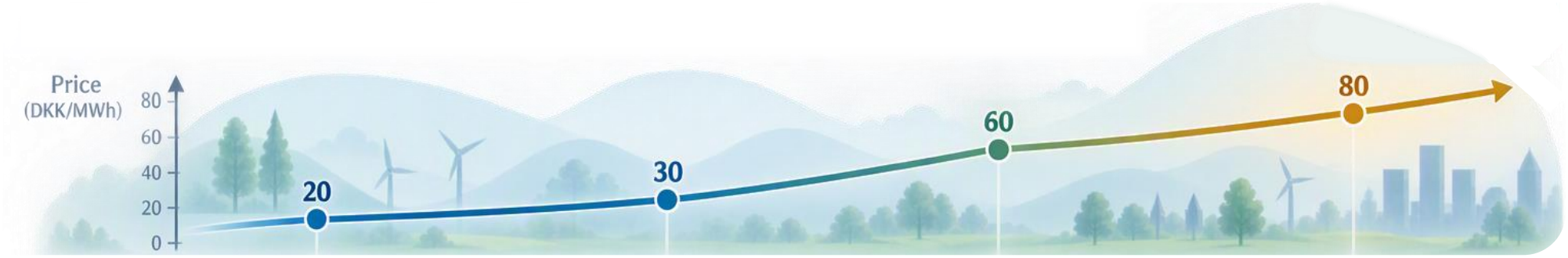
Best price spread (h1: 20 DKK/kWh → h4: 80 DKK/kWh)

Value of inter-temporal arbitrage: 80 - 20 = 60 DKK/kWh

✓ **Rank hours** from highest to lowest price spread: value for inter-temporal arbitrage

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



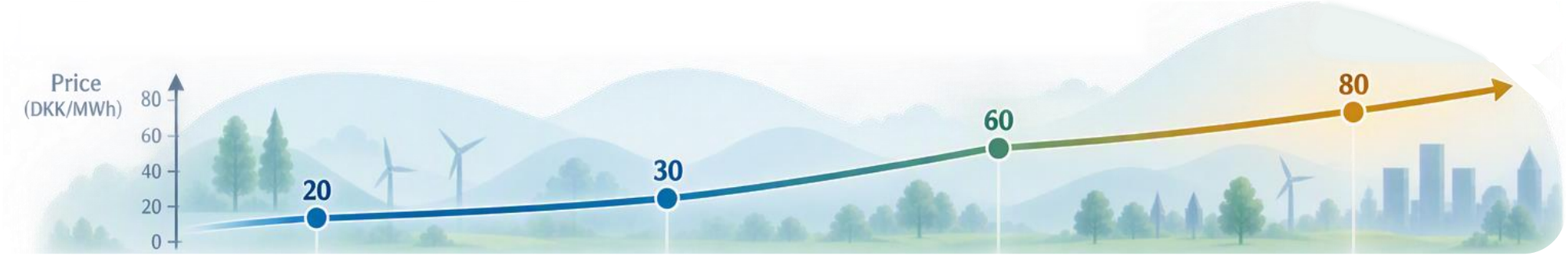
Next-best price spread (h2: 30 DKK/kWh → h3: 60 DKK/kWh)

Value of inter-temporal arbitrage: 60 - 30 = 30 DKK/kWh

✓ **Rank hours** from highest to lowest price spread: value for inter-temporal arbitrage

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



Hour 1 (t = 1)

Price $\lambda_1 = 20$ DKK/MWh

Charge
$P_1^{ch} = 10$ kW
(maximum)

Charge maximum power

Hour 4 (t = 4)

Price $\lambda_4 = 80$ DKK/MWh

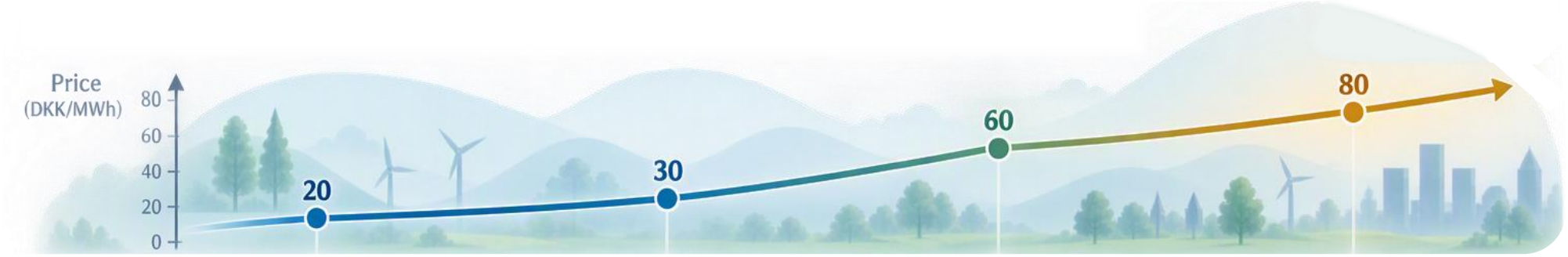
Discharge
$P_4^{dis} = 10$ kW
(maximum)

Discharge maximum power

✓ In hours with best price spread: charge-discharge as much as possible **(power limit)**

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



Hour 1 (t = 1)

Price λ_1 = 20 DKK/MWh

Charge
P_1^{ch} = 10 kW
(maximum)

Energy stored
10 kWh

Charge maximum power

Hour 4 (t = 4)

Price λ_4 = 80 DKK/MWh

Discharge
P_4^{dis} = 10 kW
(maximum)

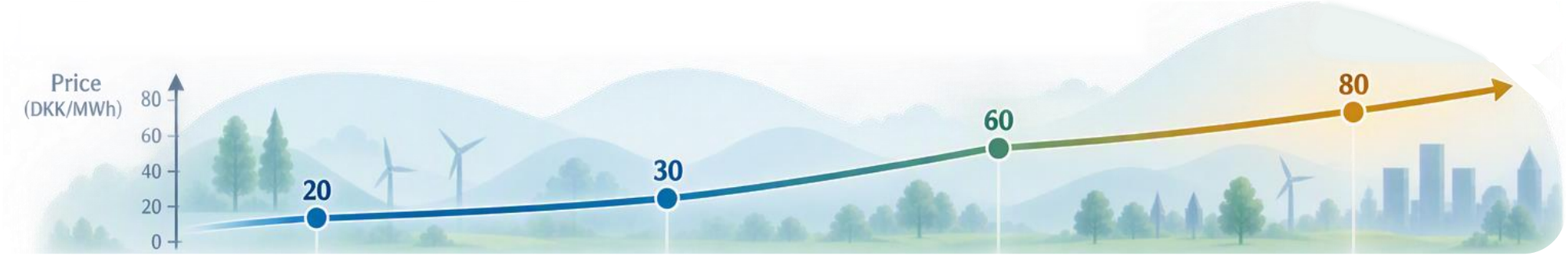
Energy stored
0 kWh
(empty)

Discharge maximum power

✓ In hours with best price spread: charge-discharge as much as possible **(power limit)**

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



Hour 1 (t = 1)

Price $\lambda_1 = 20$ DKK/MWh

Charge $P_1^{ch} = 10$ kW (maximum)

Energy stored 10 kWh

Charge maximum power

Hour 2 (t = 2)

Price $\lambda_2 = 30$ DKK/MWh

Charge $P_2^{ch} = 5$ kW

Energy stored 15 kWh (full)

Charge remaining energy

Hour 4 (t = 4)

Price $\lambda_4 = 80$ DKK/MWh

Discharge $P_4^{dis} = 10$ kW (maximum)

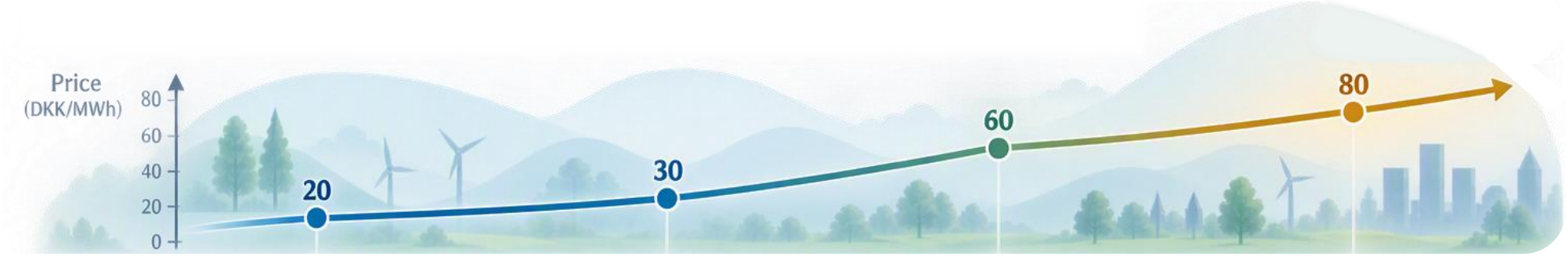
Energy stored 0 kWh (empty)

Discharge maximum power

✓ In hours with next-best spread: charge-discharge **remaining energy (linked across time)**

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



Hour 1 (t = 1)

Price λ_1 = 20 DKK/MWh

Charge P_1^{ch} = 10 kW (maximum)

Energy stored 10 kWh

Charge maximum power

Hour 2 (t = 2)

Price λ_2 = 30 DKK/MWh

Charge P_2^{ch} = 5 kW

Energy stored 15 kWh (full)

Charge remaining energy

Hour 3 (t = 3)

Price λ_3 = 60 DKK/MWh

Discharge P_3^{dis} = 5 kW

Energy stored 10 kWh

Discharge remaining energy

Hour 4 (t = 4)

Price λ_4 = 80 DKK/MWh

Discharge P_4^{dis} = 10 kW (maximum)

Energy stored 0 kWh (empty)

Discharge maximum power

✓ In hours with next-best spread: charge-discharge **remaining energy (linked across time)**

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



Questions:

Why not discharge more at hour 3?

Hourly price is > average daily price and no capacity/energy constraint is binding in this hour

Four-hour toy example: Solutions

Merit order for charging/discharging combinations follows decreasing price spreads



Questions:

Why not discharge more at hour 3?

Hourly price is > average daily price and no capacity/energy constraint is binding in this hour



Observations:

- Battery keeps 10 kWh to discharge in hour 4 (at 80 DKK/kWh)
- **Inter-temporal opportunity cost**: Energy sold at 60 DKK/kWh is energy that cannot be sold later at 80 DKK/kWh

Self-scheduling problem: decisions, objective, constraints

Input parameters

E_0: initial state of charge (SoC) [kWh]

$0 < \rho^{ch} < 1$: charging efficiency

$0 < \rho^{dis} < 1$: discharging efficiency

$\overline{P}^{ch}$: maximum charging power [kW]

$\overline{P}^{dis}$: maximum charging power [kW]

$\overline{E}$: maximum SoC [kWh]

$\lambda_t > 0$: energy price at hour t [DKK/kWh]

Decision variables

$p_t^{ch} \geq 0$:charging power at hour t [kW]

$p_t^{dis} \geq 0$:discharging power at hour t [kW]

$e_t \geq 0$:SoC at the end of hour t [kWh]

Self-scheduling problem: decisions, objective, constraints

Input parameters

E_0: initial state of charge (SoC) [kWh]

$0 < \rho^{ch} < 1$: charging efficiency

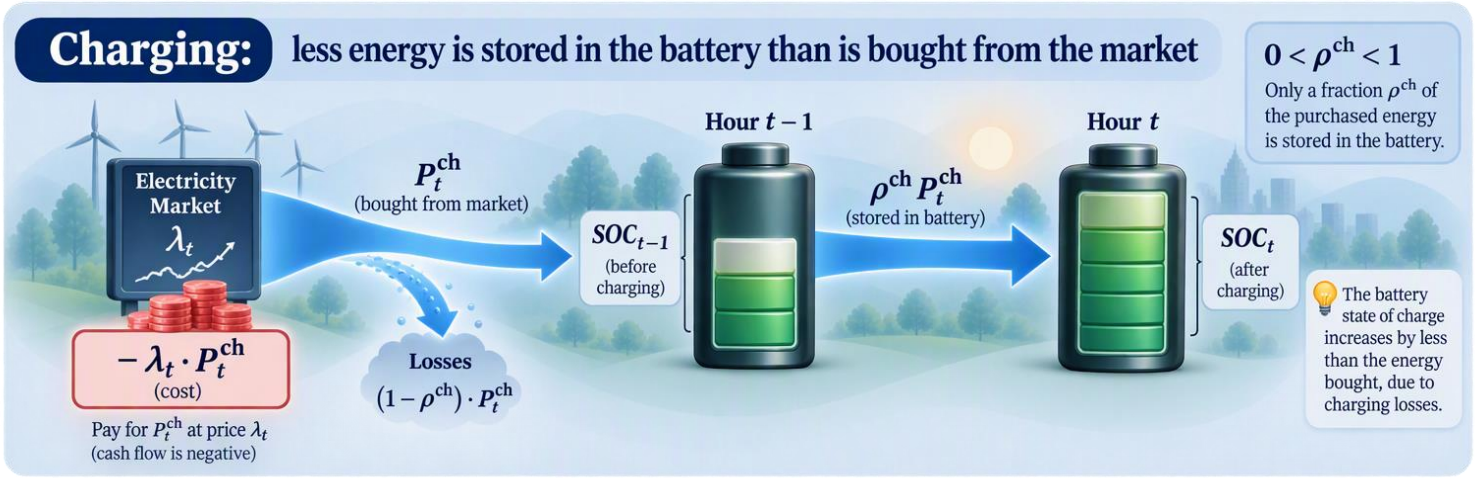
$0 < \rho^{dis} < 1$: discharging efficiency

$\overline{P}^{ch}$: maximum charging power [kW]

$\overline{P}^{dis}$: maximum charging power [kW]

$\overline{E}$: maximum SoC [kWh]

$\lambda_t > 0$: energy price at hour t [DKK/kWh]



Decision variables

$p_t^{ch} \geq 0$:charging power at hour t [kW]

$p_t^{dis} \geq 0$:discharging power at hour t [kW]

$e_t \geq 0$:SoC at the end of hour t [kWh]

Self-scheduling problem: decisions, objective, constraints

Input parameters

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$0 < \rho^{dis} < 1$: discharging efficiency

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$\overline{P}^{dis}$: maximum charging power [kW]

$\overline{E}$: maximum SoC [kWh]

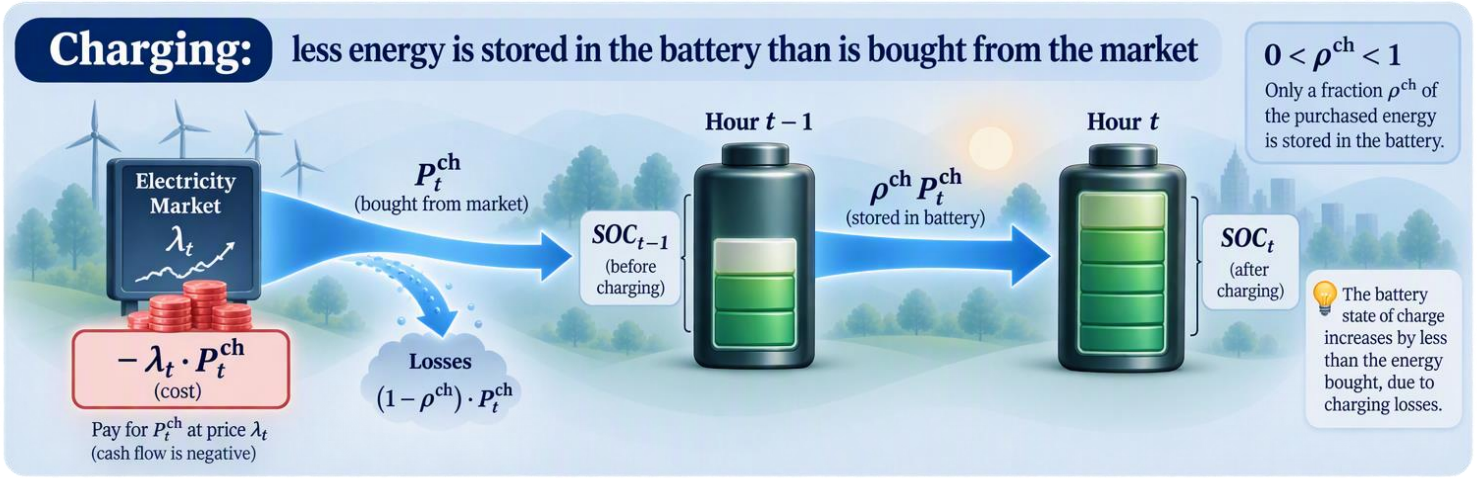
$\lambda_t > 0$: energy price at hour t [DKK/kWh]

Decision variables

$p_t^{ch} \geq 0$:charging power at hour t [kW]

$p_t^{dis} \geq 0$:discharging power at hour t [kW]

$e_t \geq 0$:SoC at the end of hour t [kWh]



Self-scheduling problem: decisions, objective, constraints

Primal formulation (intermediate)

$$\max_{p^{ch},p^{dis},e} \quad \sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right) \qquad (\mathcal{P}_0)$$

Profits = cash flows from energy sold – energy purchased

ψ_t :multiplier of the hourly energy balance at hour t (DKK/kWh) · $\underline{\delta}_t, \overline{\delta}_t \geq 0$:multipliers of the energy bounds at hour t (DKK/kWh) · $\underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}, \underline{\mu}_t^{dis}, \overline{\mu}_t^{dis} \geq 0$:multipliers of the power bounds at hour t (DKK/kW) · All other notations are the same as in Slide 18

Self-scheduling problem: decisions, objective, constraints

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Profits = cash flows from energy sold – energy purchased

s.t.

$$e_1 = E_0 + \rho^{ch} p_1^{ch} - \frac{1}{\rho^{dis}} p_1^{dis} \qquad (t = 1) \qquad : \psi_1 \qquad (\mathcal{P}_1)$$

Initial SoC dynamics

$$e_t = e_{t-1} + \rho^{ch} p_t^{ch} - \frac{1}{\rho^{dis}} p_t^{dis}, \qquad \forall t > 1 \qquad : \psi_t \qquad (\mathcal{P}_1)$$

SoC dynamics

ψ_t :multiplier of the hourly energy balance at hour t (DKK/kWh) · $\underline{\delta}_t, \overline{\delta}_t \geq 0$:multipliers of the energy bounds at hour t (DKK/kWh) · $\underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}, \underline{\mu}_t^{dis}, \overline{\mu}_t^{dis} \geq 0$:multipliers of the power bounds at hour t (DKK/kW) · All other notations are the same as in Slide 18

Self-scheduling problem: decisions, objective, constraints

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Initial SoC dynamics

$$e_t = e_{t-1} + \rho^{ch} p_t^{ch} - \frac{1}{\rho^{dis}} p_t^{dis}, \qquad \forall t > 1 \qquad : \psi_t \qquad (\mathcal{P}_1)$$

SoC dynamics

$$0 \le p_t^{ch} \le \overline{P}^{ch}, \qquad \forall t \qquad : \underline{\mu}_t^{ch}, \overline{\mu}_t^{ch} \qquad (\mathcal{P}_2), (\mathcal{P}_3)$$

Charging power bounds

$$0 \le p_t^{dis} \le \overline{P}^{dis}, \qquad \forall t \qquad : \underline{\mu}_t^{dis}, \overline{\mu}_t^{dis} \qquad (\mathcal{P}_4), (\mathcal{P}_5)$$

Discharging power bounds

ψ_t :multiplier of the hourly energy balance at hour t (DKK/kWh) · $\underline{\delta}_t, \overline{\delta}_t \ge 0$:multipliers of the energy bounds at hour t (DKK/kWh) · $\underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}, \underline{\mu}_t^{dis}, \overline{\mu}_t^{dis} \ge 0$:multipliers of the power bounds at hour t (DKK/kW) · All other notations are the same as in Slide 18

Self-scheduling problem: decisions, objective, constraints

Primal formulation (intermediate)

$\max_{p^{ch},p^{dis},e}$	$\sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$			$(\mathcal{P}_0)$	Profits = cash flows from energy sold – energy purchased
s.t.	$e_1 = E_0 + \rho^{ch} p_1^{ch} - \frac{1}{\rho^{dis}} p_1^{dis}$	$(t=1)$	$:\psi_1$	$(\mathcal{P}_1)$	Initial SoC dynamics
	$e_t = e_{t-1} + \rho^{ch} p_t^{ch} - \frac{1}{\rho^{dis}} p_t^{dis},$	$\forall t > 1$	$:\psi_t$	$(\mathcal{P}_1)$	SoC dynamics
	$0 \le p_t^{ch} \le \overline{P}^{ch},$	$\forall t$	$:\underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}$	$(\mathcal{P}_2), (\mathcal{P}_3)$	Charging power bounds
	$0 \le p_t^{dis} \le \overline{P}^{dis},$	$\forall t$	$:\underline{\mu}_t^{dis}, \overline{\mu}_t^{dis}$	$(\mathcal{P}_4), (\mathcal{P}_5)$	Discharging power bounds
	$0 \le e_t \le \overline{E},$	$\forall t$	$:\underline{\delta}_t, \overline{\delta}_t$	$(\mathcal{P}_6), (\mathcal{P}_7)$	Energy bounds

ψ_t :multiplier of the hourly energy balance at hour t (DKK/kWh) · $\underline{\delta}_t, \overline{\delta}_t \ge 0$:multipliers of the energy bounds at hour t (DKK/kWh) · $\underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}, \underline{\mu}_t^{dis}, \overline{\mu}_t^{dis} \ge 0$:multipliers of the power bounds at hour t (DKK/kW) · All other notations are the same as in Slide 18

Self-scheduling problem: decisions, objective, constraints

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	$0 \le e_t \le \overline{E},$	$\forall t$	$:\underline{\delta}_t, \overline{\delta}_t$	$(\mathcal{P}_6), (\mathcal{P}_7)$	Energy bounds

Structure: LP + weak slater condition→ strong duality holds, KKTs necessary & sufficient

ψ_t :multiplier of the hourly energy balance at hour t (DKK/kWh) · $\underline{\delta}_t, \overline{\delta}_t \ge 0$:multipliers of the energy bounds at hour t (DKK/kWh) · $\underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}, \underline{\mu}_t^{dis}, \overline{\mu}_t^{dis} \ge 0$:multipliers of the power bounds at hour t (DKK/kW) · All other notations are the same as in Slide 18



With this model formulation, what prevents the battery from charging and discharging simultaneously?

- ▲ The state-of-charge balance equation
- ◆ Efficiency losses and positive prices make it suboptimal
- ● Nothing - unless binary variables are introduced



With this model formulation, if the battery starts full, at the end of the day, it will be:

- ▲ Empty - it sold everything
- ◆ It depends on tomorrow's average prices
- ● Full - it always sells as much as what it bought

Adding an end-of-horizon condition: Minimum energy

and observing impact on the schedule and profits

Primal formulation (intermediate)

$\max_{p^{ch},p^{dis},e}$	$\sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$			$(\mathcal{P}_0)$	Profits = cash flows from energy sold – energy purchased
s.t.	$e_1 = E_0 + \rho^{ch} p_1^{ch} - \frac{1}{\rho^{dis}} p_1^{dis}$	$(t=1)$	$:\psi_1$	$(\mathcal{P}_1)$	Initial SoC dynamics
	$e_t = e_{t-1} + \rho^{ch} p_t^{ch} - \frac{1}{\rho^{dis}} p_t^{dis},$	$\forall t > 1$	$:\psi_t$	$(\mathcal{P}_1)$	SoC dynamics
	$0 \le p_t^{ch} \le \overline{P}^{ch},$	$\forall t$	$:\underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}$	$(\mathcal{P}_2), (\mathcal{P}_3)$	Charging power bounds
	$0 \le p_t^{dis} \le \overline{P}^{dis},$	$\forall t$	$:\underline{\mu}_t^{dis}, \overline{\mu}_t^{dis}$	$(\mathcal{P}_4), (\mathcal{P}_5)$	Discharging power bounds
	$0 \le e_t \le \overline{E},$	$\forall t < T$	$:\underline{\delta}_t, \overline{\delta}_t$	$(\mathcal{P}_6), (\mathcal{P}_7)$	Energy bounds

$\underline{E}^{end}$:minimum energy stored at end of horizon (kWh) · All other notations are the same as in previous slides

Adding an end-of-horizon condition: Minimum energy

and observing impact on the schedule and profits

Primal formulation (final)

$\max_{p^{ch},p^{dis},e}$	$\sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$			$(\mathcal{P}_0)$	Profits = cash flows from energy sold – energy purchased
s.t.	$e_1 = E_0 + \rho^{ch} p_1^{ch} - \frac{1}{\rho^{dis}} p_1^{dis}$	$(t = 1)$	$: \psi_1$	$(\mathcal{P}_1)$	Initial SoC dynamics
	$e_t = e_{t-1} + \rho^{ch} p_t^{ch} - \frac{1}{\rho^{dis}} p_t^{dis},$	$\forall t > 1$	$: \psi_t$	$(\mathcal{P}_1)$	SoC dynamics
	$0 \le p_t^{ch} \le \overline{P}^{ch},$	$\forall t$	$: \underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}$	$(\mathcal{P}_2), (\mathcal{P}_3)$	Charging power bounds
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	$0 \le e_t \le \overline{E},$	$\forall t < T$	$: \underline{\delta}_t, \overline{\delta}_t$	$(\mathcal{P}_6), (\mathcal{P}_7)$	Energy bounds
	$\underline{E}^{end} \le e_T \le \overline{E}$	$(t = T)$	$: \underline{\delta}_T, \overline{\delta}_T$	$(\mathcal{P}_6), (\mathcal{P}_7)$	End-of-horizon conditions

$\underline{E}^{end}$:minimum energy stored at end of horizon (kWh) · All other notations are the same as in previous slides



With this formulation of the end-of-horizon condition, if the battery starts full, at the end of day 1 it will be:

- ▲ Full - it always sells exactly as much as it buys
- ◆ Equal to the minimum energy requirement, if today's prices are positive
- ● Equal to the minimum energy requirement, regardless of today's
- ■ It depends on tomorrow's prices

Adding an end-of-horizon condition: Minimum energy

and observing impact on the schedule and profits

Optimal actions depending on end-of-day energy level
(starts full, no losses)

$\underline{E}^{end}$ = required end level	Hour 3	Hour 4
$\underline{E}^{end}$ = 0 kWh	sell 5 kWh	sell 10 kWh
$\underline{E}^{end}$ = 5 kWh	—	sell 10 kWh
$\underline{E}^{end}$ = 10 kWh	—	sell 5 kWh
$\underline{E}^{end}$ = 15 kWh	—	—

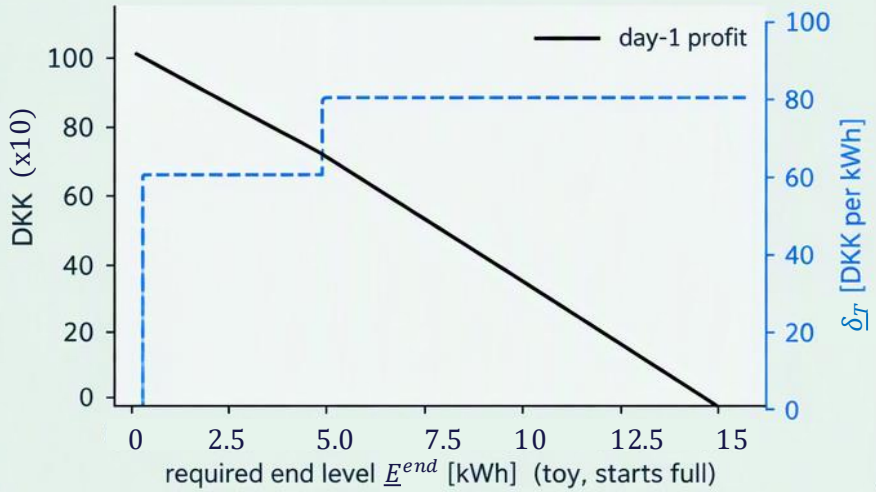
Adding an end-of-horizon condition: Minimum energy

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$\underline{E}^{end}$ = 15 kWh	—	—

Day 1 alone:



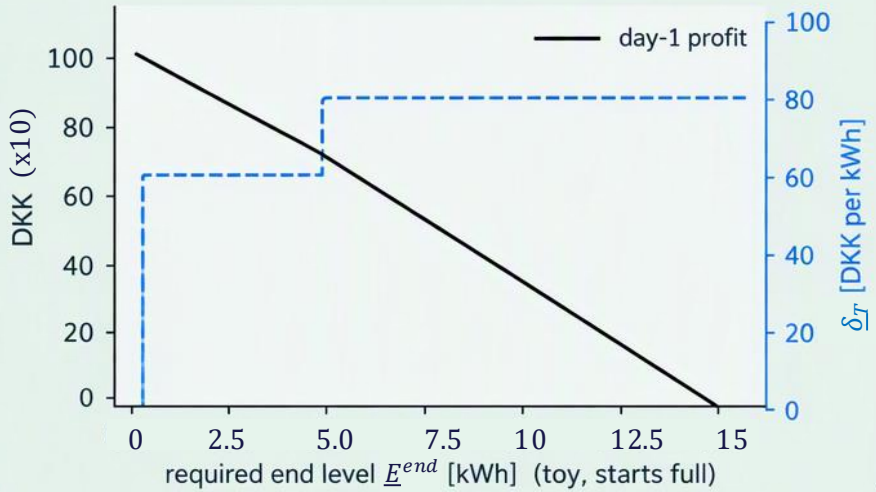
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Optimal actions depending on end-of-day energy level
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$\underline{E}^{end}$ = 10 kWh	—	sell 5 kWh
$\underline{E}^{end}$ = 15 kWh	—	—

Day 1 alone:



Insights



- Each kWh kept costs what it would have sold for during the day: 60 for the first 5 kWh (hour 3), 80 after (hour 4)
- That slope is the multiplier $\underline{\delta_T}$: the value the model puts on the end of horizon energy
- That value is **myopic** to the prices of energy the next day

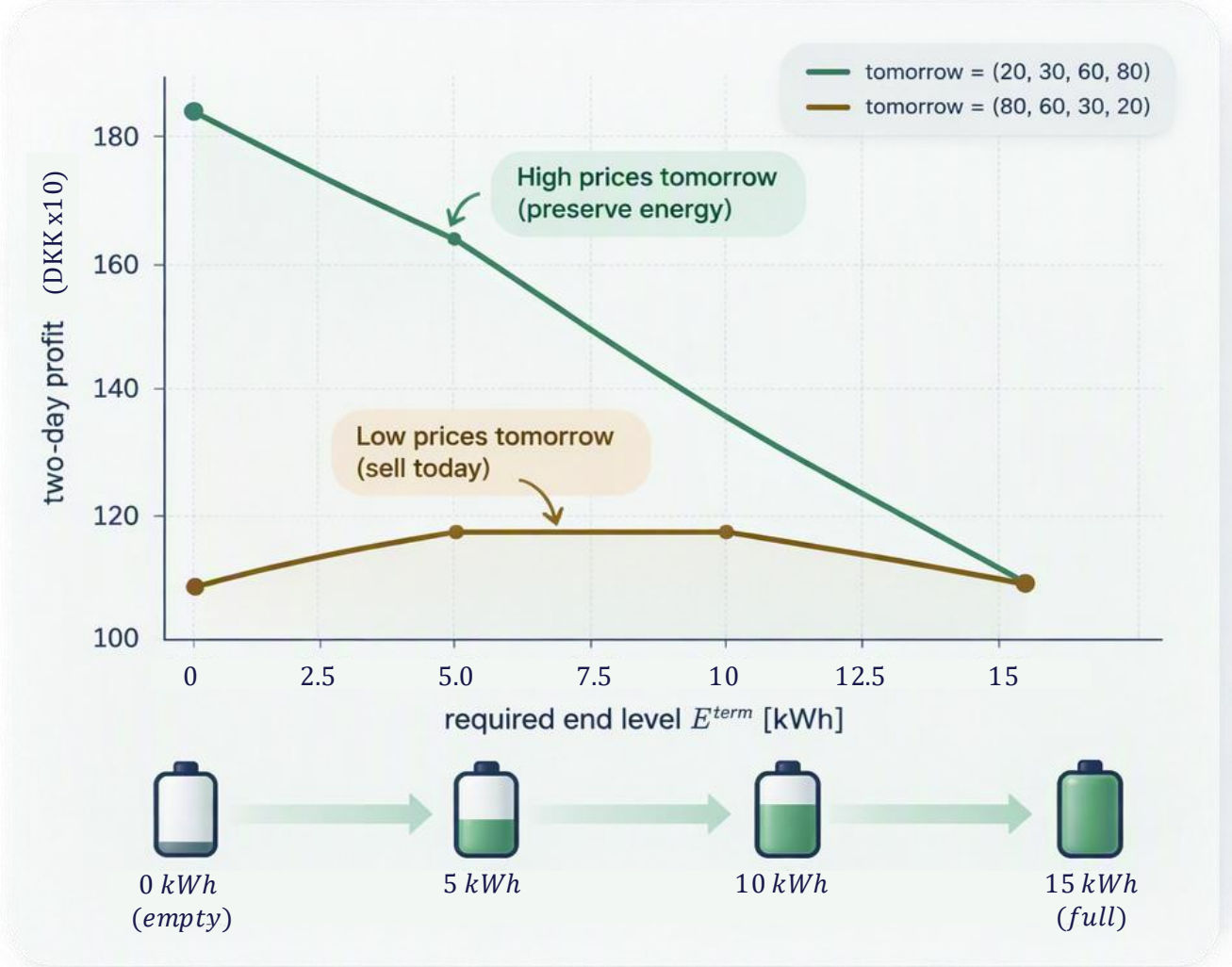


With this formulation of the end-of-horizon condition, the battery's profits over day 1 + day 2 will be...

- Always greater than profits without end-of-horizon requirements in day 1
- Always lower than profits without end-of-horizon requirements in day 1
- Neither - depends on value of minimum requirement and prices in day 2

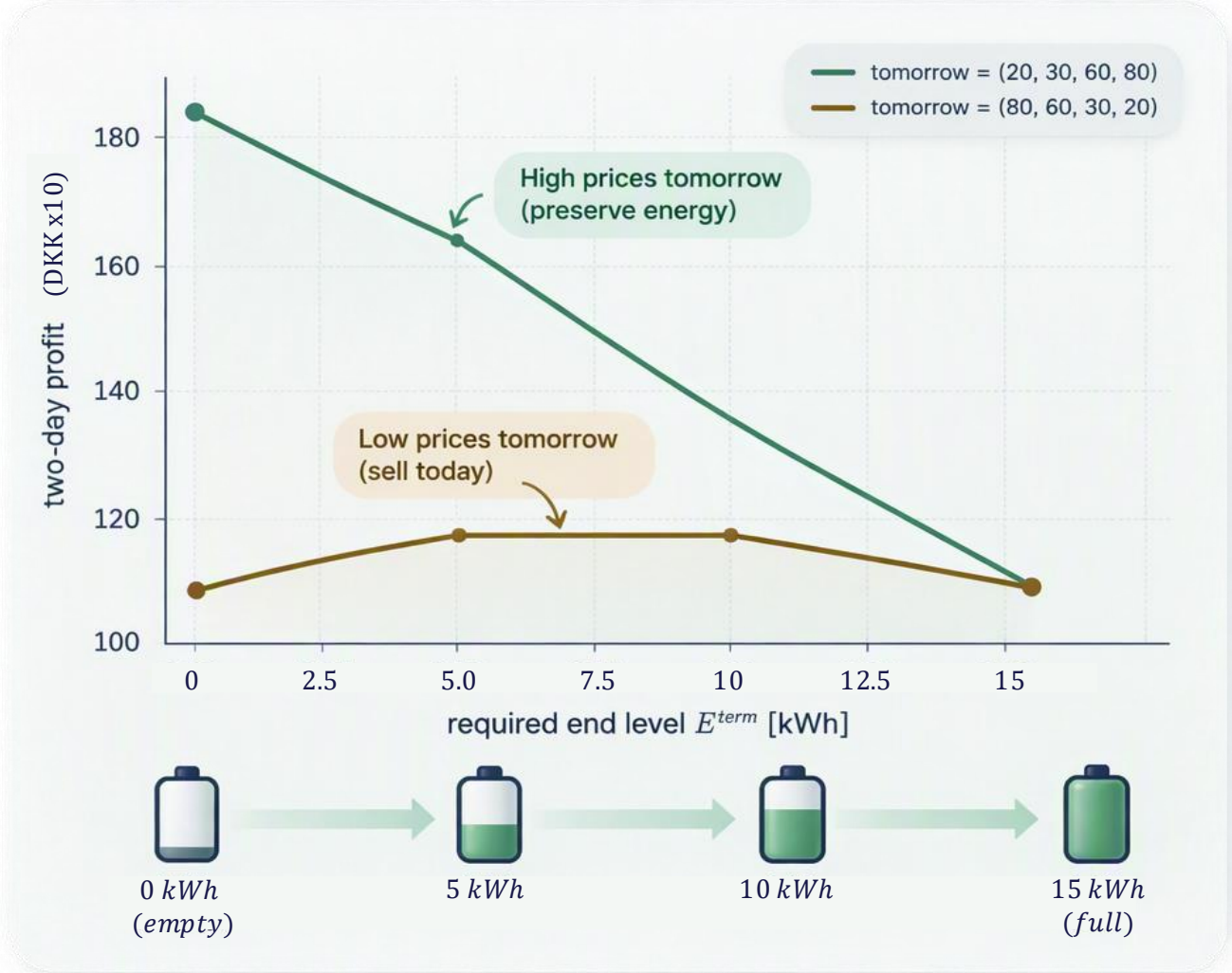
Adding an end-of-horizon condition: Minimum energy

and observing impact on the schedule and profits



Adding an end-of-horizon condition: Minimum energy

and observing impact on the schedule and profits





Choosing an end-of-horizon formulation

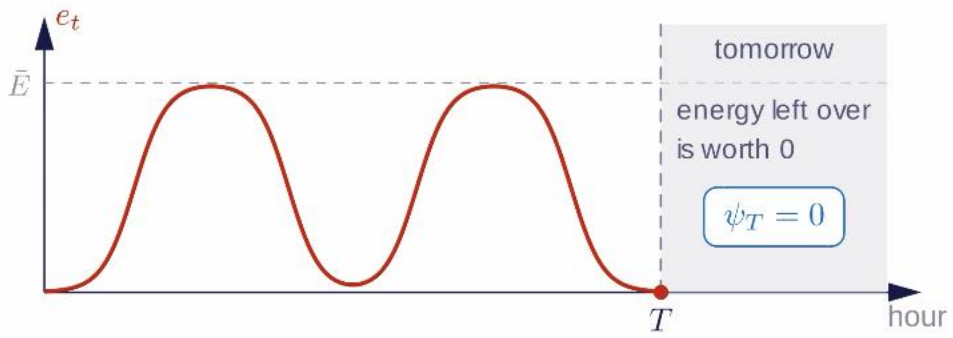
See details in Lecture notes and optional homework

Modeling choice	Guarantee at T	Information used	Sets value of stored energy to	Main risk
Free end $(\underline{E}^{end} = 0)$	none	none (worth 0)	0	dumps at the end
Minimum level $(e_T \geq \underline{E}^{end})$	yes	a daily pattern	Lagrange multiplier – revealed	myopic; infeasibility

Free end
nothing is said about the energy left at T

$\underline{E}^{end} = 0$

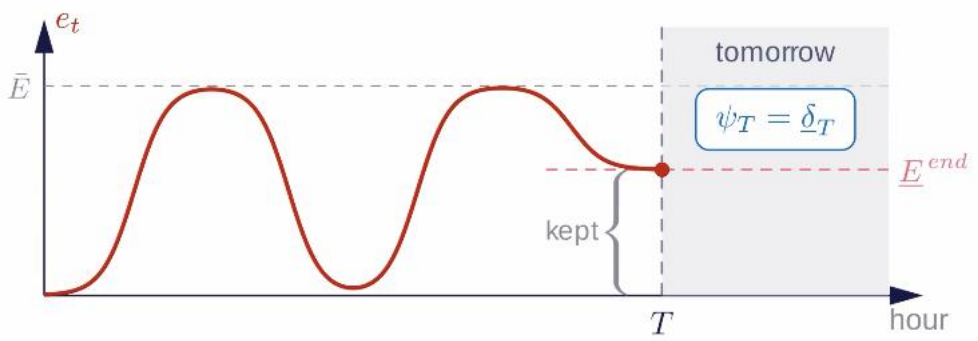
the schedule empties the battery before T



Minimum end level
a required level at T, priced by the LP

$e_T \geq \underline{E}^{end} : \underline{\delta}_T$
revealed, not chosen

the LP prices the requirement: every kWh kept has a cost



Choosing an end-of-horizon formulation

See details in Lecture notes and optional homework

Modeling choice	Guarantee at T	Information used	Sets value of stored energy to	Main risk
Free end $(\underline{E}^{end} = 0)$	none	none (worth 0)	0	dumps at the end
Minimum level $(e_T \geq \underline{E}^{end})$	yes	a daily pattern	Lagrange multiplier – revealed	myopic; infeasibility
Rolling horizon (optimize over kT, apply over T)	**no**	**a forecast of kT hours**	**Lagrange multiplier – revealed (from forecast)**	**forecast error**

Choosing an end-of-horizon formulation

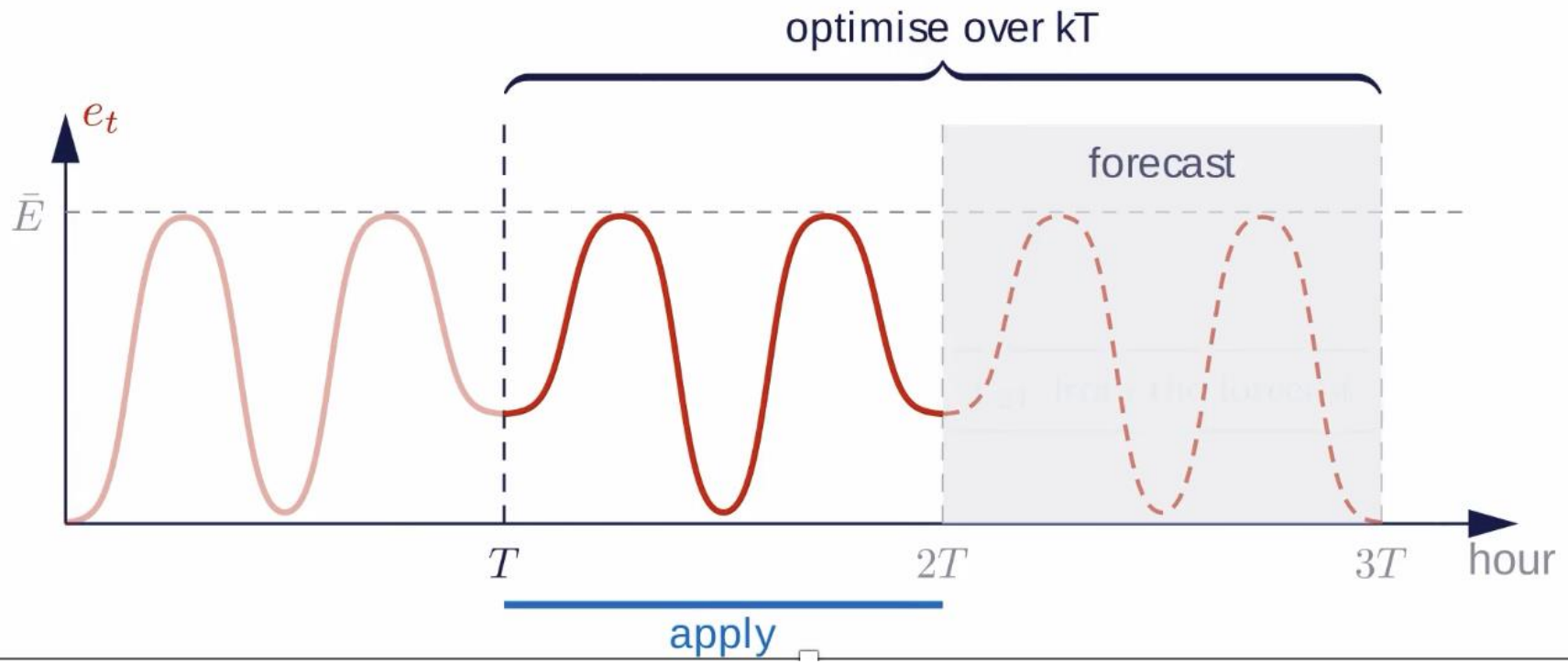
See details in Lecture notes and optional homework

Rolling horizon

optimise over kT, apply the first T, roll

$$\max \sum_{t=1}^{kT} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

hours after T are decided again tomorrow with better information



Choosing an end-of-horizon formulation

See details in Lecture notes and optional homework

Modeling choice	Guarantee at T	Information used	Sets value of stored energy to	Main risk
Free end $(\underline{E}^{end} = 0)$	none	none (worth 0)	0	dumps at the end
Minimum level $(e_T \geq \underline{E}^{end})$	yes	a daily pattern	Lagrange multiplier – revealed	myopic; infeasibility
Rolling horizon (optimize over kT, apply over T)	no	a forecast of kT hours	Lagrange multiplier – revealed (from forecast)	forecast error
Terminal value $(+\pi\, e_T)$	**no**	**one price**	**π, chosen**	**calibration**

Choosing an end-of-horizon formulation

See details in Lecture notes and optional homework

Terminal value

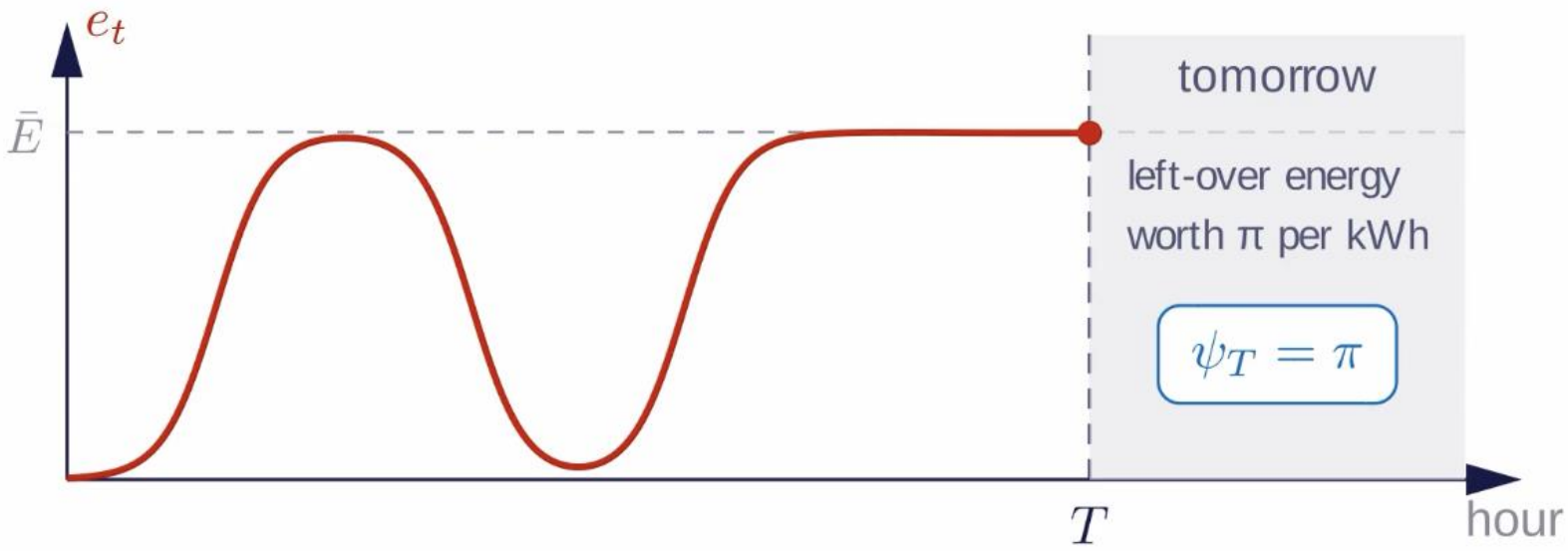
a chosen price π on the energy left at T

$$\max \sum_t \lambda_t \left(p_t^{dis} - p_t^{ch}\right) + \pi\, e_T$$

π ——●—— chosen

only as good as its calibration

keeps the kWh it could only sell below π — the higher π, the more it keeps



Choosing an end-of-horizon formulation

See details in Lecture notes and optional homework

Modeling choice	Guarantee at T	Information used	Sets value of stored energy to	Main risk
Free end $(\underline{E}^{end} = 0)$	none	none (worth 0)	0	dumps at the end
Minimum level $(e_T \geq \underline{E}^{end})$	yes	a daily pattern	Lagrange multiplier – revealed	myopic; infeasibility
Rolling horizon (optimize over kT, apply over T)	no	a forecast of kT hours	Lagrange multiplier – revealed (from forecast)	forecast error
Terminal value $(+\pi\, e_T)$	no	one price	π, chosen	calibration



- ✓ Rule of thumb: rolling horizon when a forecast and the computation are available; terminal value when speed matters and a long-horizon model can calibrate it; a minimum end level when feasibility after T is non-negotiable.
- ✓ Three metrics to compare them (see dashboard tab C2 – optional homework): two-day regret; terminal SoC deviation; the implicit terminal price of energy (against a benchmark)

Family of intertemporal constraints and costs

See details in lecture notes

Energy storage (LP)

- Charging & discharging decisions link state-of-charge in consecutive hours
- End-of-horizon constraints on state-of-charge couple decisions in all hours



Family of intertemporal constraints and costs

See details in lecture notes

Energy storage (LP)

- Charging & discharging decisions link state-of-charge in consecutive hours
- End-of-horizon constraints on state-of-charge couple decisions in all hours



Ramping rates (LP)

Maximum ramping rates couple production decisions in consecutive hours:

$$-\overline{R}^{down} \leq p_t - p_{t-1} \leq \overline{R}^{up}$$



Family of intertemporal constraints and costs

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Flexible load with daily demand (LP)

Minimum daily demand couples consumption decisions in all hours (energy not consumed now must be consumed later):

$$\sum_{t=1}^{T} p_t^L \geq \underline{E}^{day}$$



Family of intertemporal constraints and costs

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Energy storage (LP)

- Charging & discharging decisions link state-of-charge in consecutive hours
- End-of-horizon constraints on state-of-charge couple decisions in all hours



Ramping rates (LP)

Maximum ramping rates couple production decisions in consecutive hours:

$$-\overline{R}^{down} \leq p_t - p_{t-1} \leq \overline{R}^{up}$$



Unit commitment (MILP)

- Start-up/shut down decisions link commitment status in consecutive hours:

$$u_t = u_{t-1} + v_t^{SU} - v_t^{SD} \geq 0$$

- Minimum up/down times couple decisions in consecutive hours



Flexible load with daily demand (LP)

Minimum daily demand couples consumption decisions in all hours (energy not consumed now must be consumed later):

$$\sum_{t=1}^{T} p_t^L \geq \underline{E}^{day}$$



BLOCK B · 25 min

What is a kWh of stored energy worth?

Lagrangian function and KKT conditions

Primal formulation (final)

$$\max_{p^{ch},p^{dis},E} \quad \sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

s.t.

$$e_1 = E_0 + \rho^{ch} p_1^{ch} - \frac{1}{\rho^{dis}} p_1^{dis} \qquad (t=1) \qquad : \psi_1$$

$$e_t = e_{t-1} + \rho^{ch} p_t^{ch} - \frac{1}{\rho^{dis}} p_t^{dis}, \qquad \forall t > 1 \qquad : \psi_t$$

$$0 \le p_t^{ch} \le \overline{P}^{ch}, \qquad \forall t \qquad : \underline{\mu}_t^{ch}, \overline{\mu}_t^{ch}$$

$$0 \le p_t^{dis} \le \overline{P}^{dis}, \qquad \forall t \qquad : \underline{\mu}_t^{dis}, \overline{\mu}_t^{dis}$$

$$0 \le e_t \le \overline{E}, \qquad \forall t < T \qquad : \underline{\delta}_t, \overline{\delta}_t$$

$$\underline{E}^{end} \le e_T \le \overline{E} \qquad (t=T) \qquad : \underline{\delta}_T, \overline{\delta}_T$$

Lagrangian function (intermediate)

$$\mathcal{L}\left(p^{ch}, p^{dis}, e, \psi, \underline{\delta}, \overline{\delta}, \underline{\mu}^{ch}, \overline{\mu}^{ch}, \underline{\mu}^{dis}, \overline{\mu}^{dis}\right) = \ldots$$

Lagrangian function and KKT conditions

Primal formulation (final)

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$$\underline{E}^{end} \le e_T \le \overline{E} \qquad (t=T) \qquad : \underline{\delta}_T, \overline{\delta}_T$$

Lagrangian function (intermediate)

$$\mathcal{L}\left(p^{ch}, p^{dis}, e, \psi, \underline{\delta}, \overline{\delta}, \underline{\mu}^{ch}, \overline{\mu}^{ch}, \underline{\mu}^{dis}, \overline{\mu}^{dis}\right)$$

$$= -\sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

Lagrangian function and KKT conditions

Primal formulation (final)

$$\max_{p^{ch},p^{dis},E} \quad \sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

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Lagrangian function (intermediate)

$$\mathcal{L}\left(p^{ch}, p^{dis}, e, \psi, \underline{\delta}, \overline{\delta}, \underline{\mu}^{ch}, \overline{\mu}^{ch}, \underline{\mu}^{dis}, \overline{\mu}^{dis}\right)$$

$$= -\sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

$$+ \psi_1 \left(e_1 - E_0 - \rho^{ch} p_1^{ch} + \frac{1}{\rho^{dis}} p_1^{dis}\right)$$

$$+ \sum_{t>1} \psi_t \left(e_t - e_{t-1} - \rho^{ch} p_t^{ch} + \frac{1}{\rho^{dis}} p_t^{dis}\right)$$

Lagrangian function and KKT conditions

Primal formulation (final)

$$\max_{p^{ch},p^{dis},E} \quad \sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

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$$\underline{E}^{end} \le e_T \le \overline{E} \qquad (t=T) \qquad : \underline{\delta}_T, \overline{\delta}_T$$

Lagrangian function (intermediate)

$$\mathcal{L}\left(p^{ch}, p^{dis}, e, \psi, \underline{\delta}, \overline{\delta}, \underline{\mu}^{ch}, \overline{\mu}^{ch}, \underline{\mu}^{dis}, \overline{\mu}^{dis}\right)$$

$$= -\sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

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$$+ \sum_{t>1} \psi_t \left(e_t - e_{t-1} - \rho^{ch} p_t^{ch} + \frac{1}{\rho^{dis}} p_t^{dis}\right)$$

$$+ \sum_{t=1}^{T} \left\{-\underline{\mu}_t^{ch} p_t^{ch} + \overline{\mu}_t^{ch} \left(p_t^{ch} - \overline{P}^{ch}\right) - \underline{\mu}_t^{dis} p_t^{dis}\right.$$

$$\left. + \overline{\mu}_t^{dis} \left(p_t^{dis} - \overline{P}^{dis}\right)\right\} + \sum_{t<T} \left\{-\underline{\delta}_t e_t + \overline{\delta}_t \left(e_t - \overline{E}\right)\right\}$$

Lagrangian function and KKT conditions

Primal formulation (final)

$$\max_{p^{ch},p^{dis},E} \quad \sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

s.t.

$$e_1 = E_0 + \rho^{ch} p_1^{ch} - \frac{1}{\rho^{dis}} p_1^{dis} \qquad (t=1) \qquad : \psi_1$$

$$e_t = e_{t-1} + \rho^{ch} p_t^{ch} - \frac{1}{\rho^{dis}} p_t^{dis}, \qquad \forall t > 1 \qquad : \psi_t$$

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$$\underline{E}^{end} \le e_T \le \overline{E} \qquad (t=T) \qquad : \underline{\delta}_T, \overline{\delta}_T$$

Lagrangian function (final)

$$\mathcal{L}\left(p^{ch}, p^{dis}, e, \psi, \underline{\delta}, \overline{\delta}, \underline{\mu}^{ch}, \overline{\mu}^{ch}, \underline{\mu}^{dis}, \overline{\mu}^{dis}\right)$$

$$= -\sum_{t=1}^{T} \lambda_t \left(p_t^{dis} - p_t^{ch}\right)$$

$$+ \psi_1 \left(e_1 - E_0 - \rho^{ch} p_1^{ch} + \frac{1}{\rho^{dis}} p_1^{dis}\right)$$

$$+ \sum_{t>1} \psi_t \left(e_t - e_{t-1} - \rho^{ch} p_t^{ch} + \frac{1}{\rho^{dis}} p_t^{dis}\right)$$

$$+ \sum_{t=1}^{T} \left\{ -\underline{\mu}_t^{ch} p_t^{ch} + \overline{\mu}_t^{ch} \left(p_t^{ch} - \overline{P}^{ch}\right) - \underline{\mu}_t^{dis} p_t^{dis} \right.$$

$$\left. + \overline{\mu}_t^{dis} \left(p_t^{dis} - \overline{P}^{dis}\right) \right\} + \sum_{t<T} \left\{ -\underline{\delta}_t e_t + \overline{\delta}_t \left(e_t - \overline{E}\right) \right\}$$

$$+ \underline{\delta}_T \left(\underline{E}^{end} - e_T\right) + \overline{\delta}_T \left(e_T - \overline{E}\right)$$

Lagrangian function and KKT conditions

Stationarity conditions

Complementarity slackness

Primal/dual feasibility

Lagrangian function and KKT conditions

Stationarity conditions

$$\frac{\partial \mathcal{L}(.)}{\partial p_t^{ch}} = \lambda_t - \rho^{ch}\psi_t - \underline{\mu}_t^{ch} + \overline{\mu}_t^{ch} = 0, \qquad \forall t \quad (\mathcal{C}1)$$

Complementarity slackness

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Stationarity conditions

$$\frac{\partial \mathcal{L}(.)}{\partial p_t^{ch}} = \lambda_t - \rho^{ch}\psi_t - \underline{\mu}_t^{ch} + \overline{\mu}_t^{ch} = 0, \qquad \forall t \quad (\mathcal{C}1)$$

$$\frac{\partial \mathcal{L}(.)}{\partial p_t^{dis}} = -\lambda_t + \frac{1}{\rho^{dis}}\psi_t - \underline{\mu}_t^{dis} + \overline{\mu}_t^{dis} = 0, \forall t \quad (\mathcal{C}2)$$

Complementarity slackness

Primal/dual feasibility

Lagrangian function and KKT conditions

Stationarity conditions

$$\frac{\partial \mathcal{L}(.)}{\partial p_t^{ch}} = \lambda_t - \rho^{ch}\psi_t - \underline{\mu}_t^{ch} + \overline{\mu}_t^{ch} = 0, \qquad \forall t \quad (\mathcal{C}1)$$

$$\frac{\partial \mathcal{L}(.)}{\partial p_t^{dis}} = -\lambda_t + \frac{1}{\rho^{dis}}\psi_t - \underline{\mu}_t^{dis} + \overline{\mu}_t^{dis} = 0, \forall t \quad (\mathcal{C}2)$$

$$\frac{\partial \mathcal{L}(.)}{\partial e_t} = \begin{cases} \psi_t - \psi_{t+1} - \underline{\delta}_t + \overline{\delta}_t = 0 & if\ t < T \\ \psi_T - \underline{\delta}_T + \overline{\delta}_T = 0 & if\ t = T \end{cases} \quad (\mathcal{C}3)$$

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Complementarity slackness

$$\underline{\mu}_t^X . p_t^X = 0 \ ; \ \overline{\mu}_t^X . \left(p_t^X - \overline{P}^X\right) = 0, \forall t, X \in \{ch, dis\} \quad (\mathcal{C}4)$$

$$\underline{\delta}_t . e_t = 0 \ ; \ \overline{\delta}_t \left(e_t - \overline{E}\right) = 0, \ \forall t < T \ ;$$

$$\underline{\delta}_T . \left(\underline{E}^{end} - e_T\right) = 0 \ ; \ \overline{\delta}_T \left(e_T - \overline{E}\right) = 0 \quad (\mathcal{C}5)$$

Primal/dual feasibility

Lagrangian function and KKT conditions

Stationarity conditions

$$\frac{\partial \mathcal{L}(.)}{\partial p_t^{ch}} = \lambda_t - \rho^{ch}\psi_t - \underline{\mu}_t^{ch} + \overline{\mu}_t^{ch} = 0, \qquad \forall t \quad (\mathcal{C}1)$$

$$\frac{\partial \mathcal{L}(.)}{\partial p_t^{dis}} = -\lambda_t + \frac{1}{\rho^{dis}}\psi_t - \underline{\mu}_t^{dis} + \overline{\mu}_t^{dis} = 0, \forall t \quad (\mathcal{C}2)$$

$$\frac{\partial \mathcal{L}(.)}{\partial e_t} = \begin{cases} \psi_t - \psi_{t+1} - \underline{\delta}_t + \overline{\delta}_t = 0 & if\ t < T \\ \psi_T - \underline{\delta}_T + \overline{\delta}_T = 0 & if\ t = T \end{cases} \quad (\mathcal{C}3)$$

Complementarity slackness

$$\underline{\mu}_t^X . p_t^X = 0 \ ; \ \overline{\mu}_t^X . \left(p_t^X - \overline{P}^X\right) = 0, \forall t, X \in \{ch, dis\} \quad (\mathcal{C}4)$$

$$\underline{\delta}_t . e_t = 0 \ ; \ \overline{\delta}_t \left(e_t - \overline{E}\right) = 0, \ \forall t < T \ ;$$

$$\underline{\delta}_T . \left(\underline{E}^{end} - e_T\right) = 0 \ ; \ \overline{\delta}_T \left(e_T - \overline{E}\right) = 0 \quad (\mathcal{C}5)$$

Primal/dual feasibility

$$\underline{\mu}_t^X, \overline{\mu}_t^X, \underline{\delta}_t, \overline{\delta}_t \geq 0 \ \ \forall t, X \in \{ch, dis\} \quad ; \quad \psi_t \ free \ \forall t \quad (\mathcal{C}6)$$

$$\left(\mathcal{P}_1\right), \ldots \left(\mathcal{P}_7\right)$$

The multipliers: ψ_t is a step function

From ($\mathcal{C}3$) and complementarity

"Congestion" across hours

$\psi_t - \psi_{t+1} - \underline{\delta}_t + \overline{\delta}_t = 0$ ($\mathcal{C}3$) and $0 < e_t < \overline{E} \Rightarrow \psi_t = \psi_{t+1}$

✓ *Value of stored energy is constant across hours while the battery is neither full nor empty*

✓ *$|\psi_t - \psi_{t+1}|$ represents cost of scarce energy storage capacity (hitting upper or lower bound)*

End of the horizon

$\psi_T - \underline{\delta}_T + \overline{\delta}_T = 0$ ($3\mathcal{C}$) and $e_t < \overline{E} \Rightarrow \psi_T = \underline{\delta}_T$

✓ *Minimum requirement $\underline{E}^{end}$ puts the price $\underline{\delta}_T$ on the last kWh of energy stored*

The multipliers: Regimes and thresholds

From ($\mathcal{C}1$), ($\mathcal{C}2$) - without losses

CHARGING $p_t^{ch} > 0$	MARGINAL	DISCHARGING $p_t^{dis} > 0$
$(\mathcal{C}_1)$: $\overline{\mu}_t^{ch} = \psi_t - \lambda_t \geq 0$	$(\mathcal{C}_1)$: $\psi_t = \lambda_t$	$(\mathcal{C}_1)$: $\overline{\mu}_t^{dis} = \lambda_t - \psi_t \geq 0$
✓ Buying price is below the value of stored energy	✓ Indifferent: the hour that sets the value of stored energy	✓ Selling price is above the value of stored energy
✓ $\psi_t - \lambda_t$: What charging one more kW earns (linked to opportunity of earnings)		✓ $\lambda_t - \psi_t$: What discharging one more kW earns (linked to missed opportunity of discharging in another hour)

Insights



Read ψ_t as an offer/bid price

A battery behaves like a generator with marginal cost ψ_t and a load with utility ψ_t - the same price on both sides.

The multipliers: Regimes and thresholds

From ($\mathcal{C}1$), ($\mathcal{C}2$) - with losses

CHARGING $p_t^{ch} > 0$

$$(\mathcal{C}_1) \Rightarrow \quad \boldsymbol{\rho^{ch}}\psi_t \geq \lambda_t$$

✓ Each kWh bought at λ_t stores ρ^{ch} kWh worth ψ_t each: buying is worth it

✓ $\boldsymbol{\rho^{ch}}\,\psi_t - \lambda_t$: What charging one more kW earns

IDLE / MARGINAL

$$(\mathcal{C}_1)\,(\mathcal{C}_2) \Rightarrow \boldsymbol{\rho^{ch}}\psi_t \leq \lambda_t \leq \frac{1}{\boldsymbol{\rho^{dis}}}\psi_t$$

✓ Idle strictly inside the band: buying costs more than the ρ^{ch} kWh stored are worth, selling earns less than the $\frac{1}{\boldsymbol{\rho^{dis}}}$ kWh taken out are worth - **the losses eat the margin**

DISCHARGING $p_t^{dis} > 0$

$$(\mathcal{C}_2) \Rightarrow \lambda_t \geq \frac{1}{\rho^{dis}}\psi_t$$

✓ Each kWh sold at λ_t takes out $\frac{1}{\boldsymbol{\rho^{dis}}}$ kWh worth ψ_t each: selling is worth it

✓ $\lambda_t - \frac{1}{\rho^{dis}}\psi_t$: What discharging one more kW earns



In the toy example, what is ψ_3 (value of stored energy after hour 3) equal to? Why?

- ▲ 20 DKK/kWh
- ◆ 30 DKK/kWh
- ● 60 DKK/kWh
- ■ 80 DKK/kWh

BLOCK C · 35 min

What drives the battery's profit?



In the toy example, if the energy capacity doubles ($\bar{E}$ = 30 kWh) with same $\bar{P}$. The profit will change from 750 DKK to...

- ▲ 750 DKK - it stays the same
- ◆ between 750 and 1500 DKK
- ● 1500 DKK - it doubles

Battery sizing & profits: Dashboard in pairs (15 min)

Dashboard tab C1 · preset C1a 'toy' then C1b '24-h class day'



! Do:

- **1.** Toy: sweep $\bar{E}$ from 0 to 30 kWh at $\bar{P}$ = 10 kW. Where does the profit stop growing, and why? Read $\Sigma_t \bar{\delta}_t$ next to the curve.
- **2.** Toy: sweep $\bar{P}$ from 0 to 25 kW at $\bar{E}$ = 15 kWh. Same questions; read $\Sigma_t (\bar{\mu}^{ch}_t + \bar{\mu}^{dis}_t)$.
- **3.** 24-h class day ($\bar{E}$ = 25 kWh, $\bar{P}$ = 10 kW): repeat sweep 1. At which $\bar{E}$ does 'profit per kWh' start to fall? Compare with the length of the cheap and expensive blocks.
- **4.** Profiles 'one evening peak' then 'square wave H = 2 h': which of $\bar{E}$ or $\bar{P}$ matters more on each — and why?

! Observe & reflect:

- the profit curve is piecewise linear; the multiplier sum is its slope and drops at each kink
- where the kink sits relative to the length of the price blocks
- short peaks reward power, long peaks reward energy
- **When does the max SoC matter most, and when the max power?**
- **what does the sum of the multipliers tells an investor?**


DTU